

Non-Invasive Blood Glucose Estimation Through Vascular Contraction Signal Analysis

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Abstract— Diabetes is a critical health concern globally, as it impacts overall health and serves as a precursor to other conditions. Regular monitoring of blood glucose levels is essential for effective diabetes management. However, current glucose monitoring devices are invasive, leading to tissue damage, pain, and an increased risk of infection. This paper presents a continued study on the development of a non-invasive blood glucose estimation system. Using transmitting light energy in the near-infrared wavelength and near-red wavelength to detect and record PPG signals using an ESP32 microcontroller with a MAX30102 sensor. The MAX30102 sensor can transmit and receive light in red (660 nm) and near-infrared (880 nm) and measure on the fingertip. PPG signals are then analyzed for their prominent features, revealing the relationship between blood glucose levels and the most prominent features, namely root mean square of successive differences (RMSSD) power in the high frequency range (PowHF) and ratio of low frequency range to high frequency range (LF/HF) of the red wavelength. Subsequently, these features were utilized using Multiple regression equations, with one and two variables. Each of 10 Subject participated experimental session lasts approximately 2 hours and 35 minutes. Finally, the best-performing model is the two-variable regression equation model for data from the second segment after signal filtering consisting of Glucose, RMSSD and PowHF of the red wavelength, achieving an accuracy of 89.86%.

Keywords— non-invasive blood glucose monitoring, blood glucose, photoplethysmogram, heart rate variability.

I. INTRODUCTION

Diabetes is one of the major non-communicable diseases that the United Nations has identified as a global public health issue, requiring strict control and cooperation from all countries. Due to the high mortality rate and the projected increase in diabetes cases to 5.3 million by 2040 [1], this disease remains a significant concern. Diabetes is often associated with obesity, which contributes to the development of insulin resistance and impairs the ability of the body's cells to effectively utilize insulin for converting glucose into energy. As a result, glucose accumulates in the bloodstream, hyperglycemia and subsequently, type 2 diabetes [2]. Excess

body fat has also been found to contribute to inflammation and oxidative stress, which further impairs insulin function and glucose metabolism [3].

The concentration of glucose in the blood is the primary criterion for diagnosing diabetes. To reduce the complications and negative health effects associated with diabetes, it is essential to employ devices capable of continuously monitoring blood glucose levels. Such devices facilitate timely interventions to maintain optimal glucose levels.

Currently, blood glucose monitoring methods fall into two categories: invasive and non-invasive. Invasive methods involve pricking the fingertip with a small needle to collect a blood sample, which is then placed on a test strip and read by a device. However, for those who need to monitor glucose levels daily, repeated finger pricking can cause pain, tissue damage, and an increased risk of infection [4]. Non-invasive methods, such as the device namely GlucoWatch Biographer, which was discontinued in 2007 [5], have paved the way for the development of continuous glucose monitoring (CGM) but CGM remain invasive, using microneedles inserted under the skin to measure blood glucose levels [6]. The disadvantage of these devices is their high cost, as they require frequent replacement.

Due to these challenges, several studies have been conducted to develop non-invasive blood glucose monitoring systems using photoplethysmography (PPG). This method involves transmitting light through areas of the body where vasoconstriction can be detected, with the reflected light captured by a sensor and converted into an electrical signal [7]. However, the sample size in the paper is insufficient to clearly establish the relationship between the PPG signal and blood glucose levels (BGLs). This study aims to develop a non-invasive blood glucose estimation system that employs non-invasive light-based measurement with red (660 nm) and near-infrared (880 nm) wavelength to detect and record PPG signals [8]. The recorded PPG signals were then processed and analyzed using a regression technique to estimate BGLs, compared with the BGLs read by the CGM device.

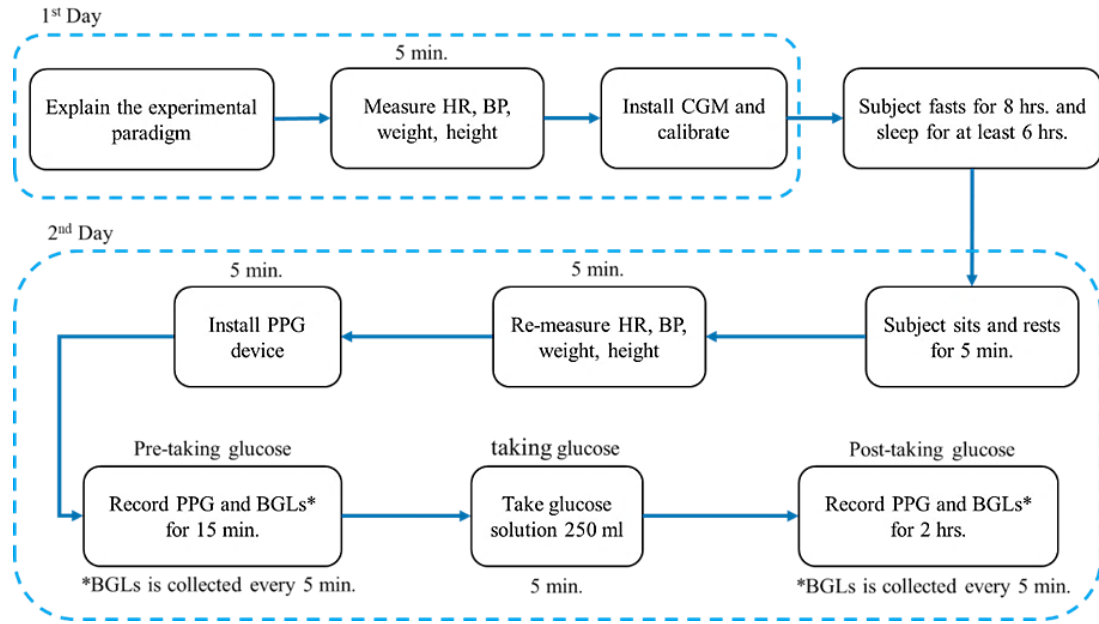


Fig. 1 The protocol for subject preparation and data collection

II. MATERIAL AND METHODS

A. Data Acquisition

In this study, an ESP32 microcontroller was employed in conjunction with a MAX30102 light sensor, which emits and receives light at red (650-670 nm) and near-infrared (870-900 nm) wavelengths. This resulted in obtained PPG signals [9]-[12] which were then transmitted from the ESP32 to a computer (using MATLAB). For monitoring BGLs, the Abbott FreeStyle Libre continuous glucose monitor was utilized [13]. Data were collected from ten healthy volunteers, comprising both male and female participants aged 18 to 27 years. PPG signals were recorded with a sampling frequency of 25 Hz.

B. Experimental Protocol for the Collection of PPG signals

The experiment was divided into two phases as illustrated in Fig. 1: subject preparation and data collection. During the subject preparation phase (1st day), healthy participants were pre-selected based on normal health parameters measurements. The continuous glucose monitoring device was attached to the screened subjects, and they were informed of the protocol, which included an eight-hour fasting period and a minimum of six hours of sleep. For the data collection phase (2nd day), each participant consumed 75 grams of glucose dissolved in 250 ml of water. The glucose is expected to be fully absorbed within two hours during a resting state [14].

Experimental protocol for the subject preparation phase (1st day) are as follows:

- 1) Explain the preliminary experimental procedure and guidelines to the participants for 5 min.
- 2) Allow the participant to rest for 5 min before measuring body composition, which includes heart rate, blood pressure, oxygen saturation (using the Schille. Argus VS-100 vital signs monitor), weight, height, and body mass index (BMI).
- 3) Install the CGM device on the participant's left upper arm, as illustrated in Fig. 2 at least 1 hr. before starting the experiment.

Experimental protocol for the data collection phase (2nd day) are as follows:

- 4) PPG acquisition device was attached on the left ring finger. PPG signal was monitored for data acquisition examination before actual data collection.
- 5) Begin recording PPG signals and collecting BGLs using the continuous glucose monitoring device. The PPG signal measurement period lasts for 15 min, with BGLs being collected every 5 min. This phase occurs before the participant consumes the glucose solution.
- 6) The participant consumes 75 grams of glucose dissolved in 250 ml of water within a five -minute period

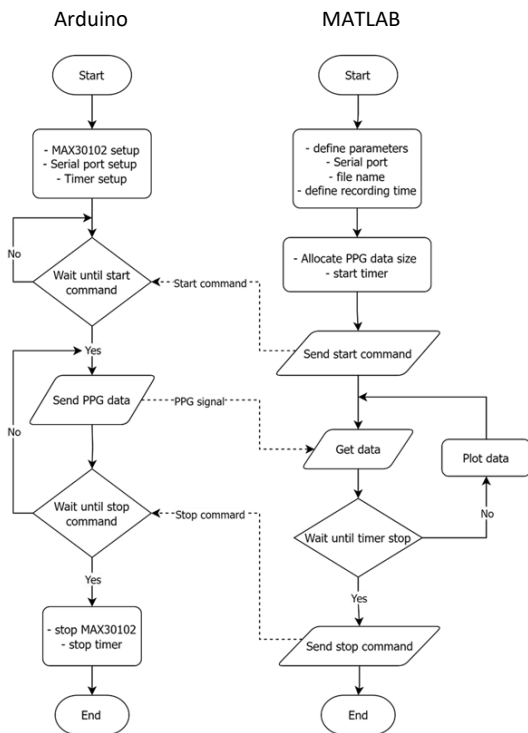


Fig. 2 Flowchart illustrating the process of collecting PPG signal.

7) Re-record the PPG signal and collect BGLs at five-minute intervals. Continue this process for two hours. During this period, the participant was instructed to remain seated in a comfortable chair.

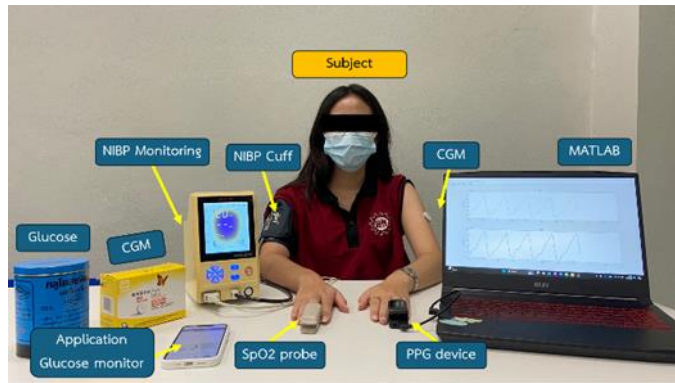


Fig. 3 An example of data acquisition devices from experiment participants.

C. Processing: Feature Extraction

The recorded PPG signals consist of two types: (i) PPG signals reflected from exposure to the near-red wavelength (PPG_{RED}) and (ii) PPG signals reflected from exposure to the near-infrared wavelength (PPG_{IR}). These PPG signals were then segmented and analyzed to extract meaningful features, which were subsequently used to determine relationship with BGLs. A design algorithm for features extraction can be described as follows [15], [16]:

- 1) Segment the raw PPG signal into 24 segments, with each segment lasting 300 s.
- 2) Apply a 6th order IIR Butterworth bandpass filter to the segmented PPG signal. (1 to 4 Hz.)
- 3) Identify the extrema (positive and negative peaks) of the filtered PPG signal using the differentiation method and estimate the Upper and Lower Envelopes using the Piecewise Cubic Hermite Interpolating Polynomial (PCHIP) method.
- 4) Calculate the time difference between positive peak-to-peak intervals (t_{ppi}) to generate a signal that represents the pulse rate variability (PRV).
- 5) Calculate the following seven features (PPG_{TSF}):

Abbreviation	Name	Description
AVNN	Average of NN Intervals	the average time between heartbeats during normal periods, where the t_{ppi} are less than 100 ms.
SDNN	Standard Deviation of NN Intervals	the standard deviation of AVNN
RMSSD	Root Mean Square of Successive Differences	the root mean square of the differences between successive Normal-to-Normal intervals
PNN50	Percentage of NN50 Intervals	the percentage of successive Normal-to-Normal intervals that differ by more than 50 ms.

Abbreviation	Name	Description
PowHF	Power in the High Frequency Range	the total power in the high-frequency range (0.15 - 0.4 Hz) of the PPG signal.
LF/HF	Ratio of Low Frequency Range to High Frequency Range	the ratio of total power in the low-frequency range to total power in the high-frequency range of the PPG signal.
MEnvS	Mean Envelope Size	The average size of the difference between the upper and lower envelope of the PPG signal.

D. Processing: Segmentation

The PPG_{TSF} dataset was segmented into three overlapping periods: (i) rising, (ii) plateau, and (iii) falling. These periods were derived from BGLs distributed over time, with each sample period being 15 min. Each period was designed to contain 10 sample points of BGLs, segmented based on individually marking the sample point at the peak glucose level as a reference point. The rising period was derived from the 10 sample points counted backward from the reference point. The plateau period combined 5 sample points counted backward from the reference point and 5 sample points counted forward from the reference point. The falling period was created from the 10 sample points counted forward from the reference point.

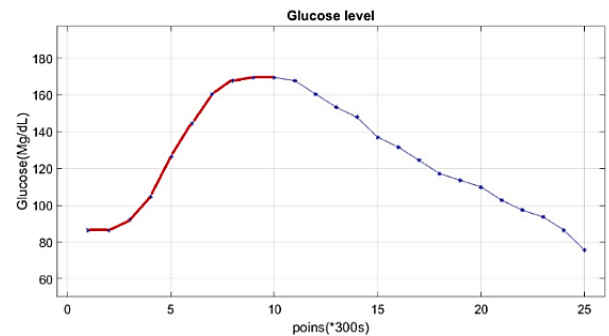


Fig. 4 Rising period: From after take glucose solution until the peak BGLs (10 data points).

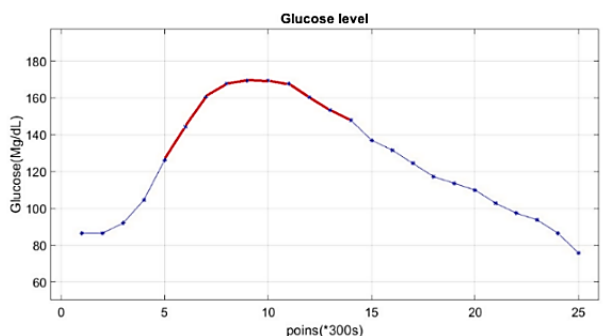


Fig. 5 Plateau period: The phase where BGLs is rising towards the peak and beginning to decline (10 data points).

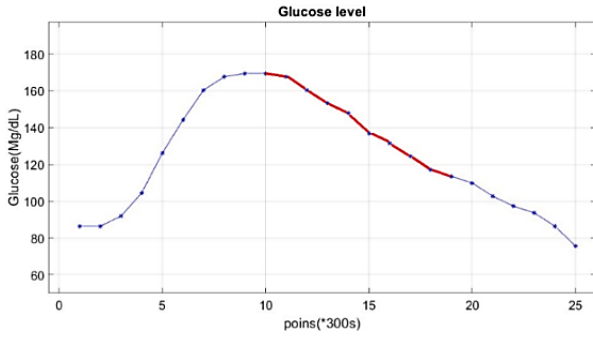


Fig. 6 Falling period: From the peak of BGLs and decline (10 data points).

These three periods are illustrated in Figure 4-6. According to period segmentation, we generated the averaged seven features to 10-sample point based on timing appearing on each 10-sample point of each period. To select features, we calculated Pearson's correlation between each feature and the collected BGLs. This was performed for all three periods.

E. Processing: Regression Model

The seven feature datasets were divided into the three periods, and a model was developed to represent a relationship between these features and blood glucose levels using a two-variable Multiple Linear Regression (MLR) equation were applied using MATLAB's Curve Fitting Tool within the polynomial model's function. Specifically, this can be explained as follows: Multiple Linear Regression [17] is a statistical method used to create a regression equation that-for appropriately predicts the value of the dependent variable, which is the variable of interest. Using multiple independent variables, also known as predictors. The prediction model created by MLR is in the form of a linear equation.

Multiple Linear Regression:

$$y = \hat{\beta}_0 + \sum_{i=1}^k \hat{\beta}_i x_i + \varepsilon \quad (1)$$

where:

- y = the dependent variable to be predicted
- x = the independent variables used as predictors
- $\hat{\beta}_0$ = y Intercept
- $\hat{\beta}_i$ = the regression coefficient for each independent variable
- ε = error term or residual*

* The error term is incorporated in other evaluation metrics, such as the root mean squared error or R-squared, to assess the accuracy of the equation.

III. RESULTS

In terms of features selection, based on the created three periods, we found that resulting top features correlated with the collected BGLs were consistent over the three periods. In Table 1, the calculated Pearson's correlation values of the top three features over the three period were illustrated. The top three selected features are RMSSD, PowHF, and LF/HF and we named them as PPG_{TSF} for comfortable use later.

Blood Glucose estimation models were created using MLR. Nine equations were created using possible combination of 2-feature input (2 variables) and 1-BGLs output as shown in Table 2. The performance of each model in estimating BGLs measured by accuracy (%), adjusted R-squared (R^2_{adj}), R-squared (R^2), and Root Mean Squared Error (RMSE) was calculated by comparing the actual BGLs measured by the standard CGM device with the estimated BGLs from the model. The results are depicted in Table 3.

TABLE I. THE CORRELATION BETWEEN BGLs AND THE PPG_{TSF} DATASET WAS ANALYZED AFTER CONSUMING A GLUCOSE SOLUTION FOR 2 HRS. THE DATASET WAS DIVIDED INTO THREE SEGMENTS

Period	RMSSD	PowHF	LF/HF
Rising	0.56	0.56	0.50
Plateau	0.50	0.63	0.50
Falling	0.71	0.53	0.47

TABLE II. THE MODELS INVOLVING 3 FEATURES FROM PPG_{TSF} FOR THREE SEGMENTS, UTILIZING TWO-VARIABLE COMPRISE A TOTAL OF 9 EQUATIONS

Period	Multiple linear regression (MLR)
Rising	$G = 87.9 + 0.1645 \text{ powhf} - 4094 \text{ rmssd}$ (2)
	$G = 80.12 + 39.2 \text{ lfhf} - 0.02098 \text{ powhf}$ (3)
	$G = 211.5 - 38.33 \text{ lfhf} - 2455 \text{ rmssd}$ (4)
Plateau	$G = 75.96 + 0.219 \text{ powhf} - 4814 \text{ rmssd}$ (5)
	$G = 28.02 + 88.6 \text{ lfhf} + 0.06105 \text{ powhf}$ (6)
	$G = 175.6 - 13.93 \text{ lfhf} - 1750 \text{ rmssd}$ (7)
Falling	$G = 100.7 + 0.1208 \text{ powhf} - 3053 \text{ rmssd}$ (8)
	$G = 23.86 + 66.02 \text{ lfhf} + 0.01846 \text{ powhf}$ (9)
	$G = 84.22 + 35.51 \text{ lfhf} - 494.9 \text{ rmssd}$ (10)

TABLE III. THE PERFORMANCE OF THE TWO-VARIABLE EQUATION MODEL FOR ESTIMATING BGLs ACROSS THE THREE DATA SEGMENTS ARE COMPARED WITH BGLs FROM CGM AND ESTIMATED BGLs FROM PPG_{TSF}

Two-variable equation					
Period	Features	Acc. (%)	R^2_{adj}	R^2	RMSE
Rising	RMSSD & PowHF	82.59	0.16	0.17	22.94
	LF/HF & PowHF	82.00	0.07	0.08	24.09
	RMSSD & LF/HF	82.43	0.13	0.13	23.40
Plateau	RMSSD & PowHF	89.86	0.44	0.45	15.65
	LF/HF & PowHF	87.26	0.19	0.19	18.94
	RMSSD & LF/HF	87.26	0.22	0.23	18.5
Falling	RMSSD & PowHF	87.37	0.23	0.24	18.64
	LF/HF & PowHF	86.87	0.19	0.20	19.09
	RMSSD & LF/HF	87.00	0.19	0.20	19.07

IV. DISCUSSION

In Table 3, we found that the most efficient blood glucose estimation model among the nine MLR models is equation (5), which achieved an accuracy of 89.86%, adjusted R-squared of 0.44, R-squared of 0.45, and Root Mean Squared Error of 15.65. Although these results indicate moderate accuracy, the current model is not yet adequate for clinical application in non-invasive blood glucose monitoring. However, the model highlights the potential of RMSSD and PowHF features, which showed the best performance in this study.

RMSSD, derived from the standard deviation of time intervals between PPG signal peaks, When the interval between PPG peaks is wide (indicating low variation of heart rhythms), BGLs tend to decrease. Conversely, when the interval between PPG peaks is narrow (indicating high variation of heart rhythms), BGLs tend to increase. However, PowHF reflects the high-frequency range (0.15 - 0.4 Hz) of the time interval between PPG peaks. This feature may imply an occurrence of high heart rate resulting from an increase in blood glucose.

V. CONCLUSION

This study involves the development of a blood glucose estimation model using a two-variable Multiple Linear Regression equation. PPG signals and BGLs were collected through a designed experimental protocol. The processing algorithm was proposed with a highlight of period segmentation and feature selection. It was found that the most suitable features extracted from the PPG signal for blood glucose estimation are RMSSD, PowHF, and LF/HF of the PPG signal reflected from exposing red wavelength. The period of plateau, the intermediate interval over the recording period, was found to be the best period for estimating model which achieves an accuracy of 89.86%. To improve accuracy of the blood glucose estimation model, additional data collection is required as well as investigation of difference PPG sensor placement such as wrist and earlobe. The findings from this study could also be considered for the future development of non-invasive blood glucose monitoring devices, which would be highly beneficial for the continuous monitoring and management of diabetes in patients. Additionally, increasing the number of participants in future studies could provide more robust data, enhancing the reliability and generalizability of the results.

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